

Experiment No. 19 To predict bank loan approval using the Naïve Bayes algorithm.

Aim

To predict bank loan approval using the **Gaussian Naïve Bayes (GaussianNB)** algorithm.

Dataset 1: Personal Loan Approval

Program

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [12,820,5],[10,790,6],[9,760,7],[7,700,8],[6,680,9],
    [5,650,10],[4,620,12],[8,730,8],[3,600,13],[11,810,5]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[9,770,6]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])
```

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [12,820,5],[10,790,6],[9,760,7],[7,700,8],[6,680,9],
    [5,650,10],[4,620,12],[8,730,8],[3,600,13],[11,810,5]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[9,770,6]]
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])
```

Output

Predicted Loan Status: Approved

Result

The loan application was successfully classified as **Approved**.

Dataset 2: Home Loan Approval

Program

```
from sklearn.naive_bayes import GaussianNB
```

```
from sklearn.model_selection import train_test_split
```

```
X = [
```

```
    [90000,850,8000],[80000,820,9000],[70000,780,12000],
```

```
    [60000,740,15000],[50000,690,18000],[45000,670,20000],
```

```
    [40000,640,23000],[75000,800,10000],[35000,620,25000],
```

```
[95000,870,7000]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[72000,790,11000]]
prediction = model.predict(new_sample)
```

```

print("Predicted Loan Status:", prediction[0])

from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [90000,850,8000],[80000,820,9000],[70000,780,12000],
    [60000,740,15000],[50000,690,18000],[45000,670,20000],
    [40000,640,23000],[75000,800,10000],[35000,620,25000],
    [95000,870,7000]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[72000,790,11000]]
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])

```

Output

Predicted Loan Status: Approved

Result

The home loan application was classified as **Approved**.

Dataset 3: Business Loan Approval

Program

```

from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [20,10,1],[18,9,1],[15,8,2],[12,6,2],[10,5,3],

```

```
[8,4,3],[6,3,4],[16,8,2],[5,2,5],[22,12,1]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[17,8,2]]
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])
```

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [20,10,1],[18,9,1],[15,8,2],[12,6,2],[10,5,3],
    [8,4,3],[6,3,4],[16,8,2],[5,2,5],[22,12,1]
]

y = [
    "Approved", "Approved", "Approved", "Approved", "Rejected",
    "Rejected", "Rejected", "Approved", "Rejected", "Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[17,8,2]]
prediction = model.predict(new_sample)

print("Predicted Loan Status:" prediction[0])
```

Output

Predicted Loan Status: Approved

Result

The business loan application was classified as **Approved**.

Dataset 4: Education Loan Approval

Program

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split
```

```
X = [
    [15,92,6],[13,89,7],[11,85,8],[9,80,9],[7,72,10],
    [6,68,11],[5,65,12],[12,87,7],[4,60,13],[16,94,5]
]
```

```
y = [
```

```
"Approved","Approved","Approved","Approved","Rejected",  
"Rejected","Rejected","Approved","Rejected","Approved"  
]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)
```

```
model = GaussianNB()  
model.fit(X_train, y_train)
```

```
new_sample = [[12,88,7]]  
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])  
  
from sklearn.naive_bayes import GaussianNB  
from sklearn.model_selection import train_test_split  
  
X = [  
    [15,92,6],[13,89,7],[11,85,8],[9,80,9],[7,72,10],  
    [6,68,11],[5,65,12],[12,87,7],[4,60,13],[16,94,5]  
]  
  
y = [  
    "Approved","Approved","Approved","Approved","Rejected",  
    "Rejected","Rejected","Approved","Rejected","Approved"  
]  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)  
  
model = GaussianNB()  
model.fit(X_train, y_train)  
  
new_sample = [[12,88,7]]  
prediction = model.predict(new_sample)  
  
print("Predicted Loan Status:", prediction[0])
```

Output

Predicted Loan Status: Approved

Result

The education loan application was classified as **Approved**.

Dataset 5: Vehicle Loan Approval

Program

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [14,3,830],[12,2.5,810],[10,2,780],[8,1.8,740],[7,1.5,690],
    [6,1.2,660],[5,1.0,630],[11,2.2,790],[4,0.8,600],[15,3.5,850]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[11,2.4,800]]
```



```
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])
```

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

X = [
    [14,3,830],[12,2.5,810],[10,2,780],[8,1.8,740],[7,1.5,690],
    [6,1.2,660],[5,1.0,630],[11,2.2,790],[4,0.8,600],[15,3.5,850]
]

y = [
    "Approved","Approved","Approved","Approved","Rejected",
    "Rejected","Rejected","Approved","Rejected","Approved"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

new_sample = [[11,2.4,800]]
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])
```

Output

Predicted Loan Status: Approved

Result

The vehicle loan application was classified as **Approved**.

Conclusion

The **Gaussian Naïve Bayes algorithm** was successfully implemented for loan approval prediction. The model was trained using numerical financial features and correctly classified new loan applications as **Approved** or **Rejected** based on learned probability distributions.